# Business Sales Performance Analytics

## 1. Project Objective

To analyze business sales data and build an interactive dashboard that helps identify revenue trends, top-performing products, profitable categories and regions, and business growth opportunities for better decision-making.

## 2. Problem Statement

The business wants to analyze its sales performance to better understand revenue trends, customer purchasing behavior, and profitability. The key business questions addressed in this project are:

1. Which products generate the highest revenue?
2. How do sales change over time?
3. Which product categories and regions are the most profitable?
4. Who are the top customers contributing to overall sales?
5. Where should the business focus its efforts to drive future growth and improve profitability?

## 3. Attribute (Column / Features):

| **Attribute Name** | **Data Type** | **Description** |
|---|---|---|
| Order Id | Text | Unique Order number |
| Order Date | Date | Date when the order was placed |
| Ship Date | Date | Date when the order was shipped |
| Ship Mode | Text | Shipping method used |
| Customer ID | Text | Unique Customer identifier |
| Customer Name | Text | Name of the customer |
| Segment | Text | Customer Segment |
| Product ID | Text | Unique product identifier |
| Category | Text | Product Category |
| Sub Category | Text | Product sub category |
| Product Name | Text | Name of the product |
| Country | Text | Country where customer |

| | | |
|---|---|---|
| City | Text | Customer city location |
| State | Text | Customer state location |
| Region | Text | Business Region (West,East,Central,South) |
| Sales | Currrency | Revenue generated from product sales |
| Quantity | Number | Number of units sold |
| Discount | Percentage | Discount provided on sales |
| Profit | Currency | Profit earned from each transaction |
| Order Year | Number | Extracted year from order date |
| Order Month | Text | Extracted month from order date |
| Shipping Days | Number | Order year - Order month |
| Discount Status | Text | Indicates whether a discount applied(discounted/No Discount) |

## 4. Tools & Technologies

- **Microsoft Excel** : Data Cleaning, Remove Duplicates,Handle Missing Values, Data Type Validation , Data Modeling (Fact & Dimension Tables)
- **Power BI :** Data Modeling, DAX Measures, Interactive Dashboards, KPI Cards, Data Visualization , Slicers & Filters
- **Git Hub :** Project Documentation, Version Control, Portfolio Showcase

## 4. Data Pre-Processing

The following data preprocessing steps were performed to ensure data quality and prepare the dataset for analysis:

### 1. Data Loading

- Imported the cleaned Excel dataset into **Power BI**.
- Verified that all tables were loaded successfully for analysis.

### 2. Duplicate Handling

- Checked for duplicate records using Excel.
- Removed duplicate rows to ensure data accuracy and consistency.

### 3. Date Format Standardization

- Standardized **Order Date** and **Ship Date** into a consistent **Date** format.
- Corrected inconsistent date formats before importing into Power BI.

### 4. Data Type Validation

Validated and assigned appropriate data types for all columns:

- **Text:** Order ID, Customer Name, Category, Product Name, Region, etc.
- **Whole Number:** Row ID, Quantity, Postal Code, Order Year.
- **Decimal Number:** Sales, Profit, Discount.
- **Date:** Order Date, Ship Date.

### 5. Feature Engineering

Created additional columns to enhance analysis:

- **Order Year** – Extracted from Order Date. = Year(Order Date)
- **Order Month** – Extracted from Order Date. =Text(Order Date,"MMM")
- **Shipping Days -** Different between date (= Order Date - Ship Date)
- **Discount Status** – Classified orders as *Discounted* or *No Discount* based on the Discount value. =IF(Discount=0,"No Discount","Discount Applied")

### 6. Data Export

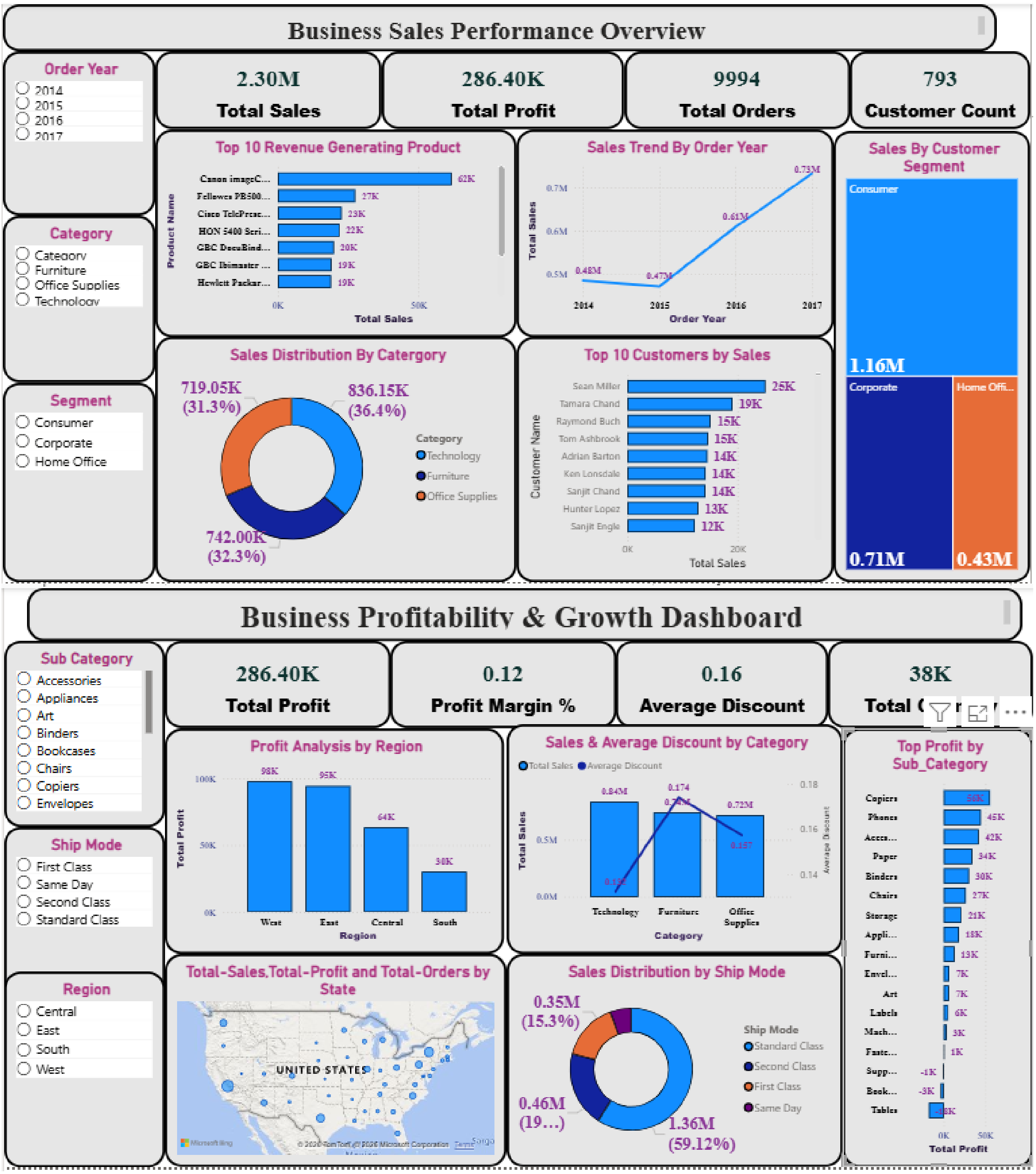
- Built a **star schema** with **Customer Dimension**, **Product Dimension**, and **Orders Fact** tables.
- Exported the cleaned dataset into CSV format for further visualization and dashboard development in Power BI.

## 5. DAX Formula(Power BI)

1. DAX Measures: Implemented DAX formulas for key metrics
2. Total Sales = SUM(Orders_Fact[Sales])
3. Total Profit = SUM(Orders_Fact[Profit])
4. Average Discount = AVERAGE(Orders_Fact[Discount])
5. Customer Count = DISTINCTCOUNT(Customer_Dim[Customer Name])
6. Profit Margin % = DIVIDE([Total Profit],[Total Sales],0)

7. Total Orders = COUNT(Orders_Fact[Order ID])
8. Total Quantity = SUM(Orders_Fact[Quantity])

## 6. Dashboard




## 7. Insights

- ➢ The business generated $2.30M in **total sales** with a **total profit** of $286.40K from 9,994 orders and 793 unique customers.
- ➢ Sales showed a consistent upward trend from **2014** to **2017**, indicating steady business growth over time.

- **Technology** emerged as the highest revenue-generating category, making it the key contributor to overall sales.
- The Consumer segment contributed the largest share of sales compared to **Corporate** and **Home Office** customers.
- The **West** region achieved the highest profitability, while the **South** region showed the lowest profit, highlighting opportunities for improvement.
- **Copiers, Phones, and Accessories** were the most profitable sub-categories, whereas Tables and Bookcases generated lower profits.
- **Standard Class** was the most frequently used shipping mode, accounting for the majority of sales.
- ·Although the business offered an average 16% discount, maintaining an average 12% profit margin indicates balanced pricing and profitability.

## 8. Recommendation

- **Increase investment in Technology products** as they generate the highest sales and have strong growth potential.
- **Improve sales performance in the South region** by introducing targeted marketing campaigns and strengthening the regional sales strategy.
- **Focus on retaining high-value customers** through loyalty programs and personalized offers to encourage repeat purchases and increase customer lifetime value.
- **Review discount policies for Furniture and other lower-profit categories** to improve profit margins while maintaining sales performance.
- **Maintain adequate inventory for top-selling products and profitable sub-categories** such as Copiers and Phones to avoid stock shortages and maximize revenue opportunities.

## 9. Conclusion

The analysis revealed consistent sales growth, strong performance from the Technology category, and significant contributions from the Consumer segment and West region. These insights enable the business to optimize sales strategies, improve profitability, and support long-term sustainable growth.